BEFORE YOU READ

The three reciprocal graphs

Cosecant, secant and cotangent are reciprocals of sine, cosine and tangent. Their graphs blow up wherever the original function is zero.

Reciprocal function graphs

csc · sec · cot

Cosecant

$\csc x = 1/\sin x$

Asymptotes

$x = n\pi$

Shape

U-branches; never in $(-1, 1)$

Secant

$\sec x = 1/\cos x$

Asymptotes

$x = \pi/2 + n\pi$

Shape

cosecant shifted by $\pi/2$

Cotangent

$\cot x = \cos x/\sin x$

Asymptotes

$x = n\pi$

Shape

strictly decreasing, period π, odd

TAKEAWAY

Asymptotes come from zeros

Every vertical asymptote of a reciprocal function sits at a zero of the function it inverts.